

Question#1 [Short Questions] [6 marks]

Q1. What is the maximum number of leaf nodes in a perfect binary tree of height 6?
(1 mark)

Answer:

Q2. Draw a Binary Tree from the given Array:

Array = [None, 7, 3, 4, 8, 2, 7, 9, None, None, None, 8, 10] (2 marks)

Answer:

Q3. getHeight(root) method returns the height of a node in a binary tree. Complete the code block. Write only the missing lines. (3 marks)

```
// Java
public int getHeight(TreeNode root) {
    if (root == null)

        1. _____

    int l = getHeight(root.left);
    int r = getHeight(root.right);

    if (l > r) {

        2. _____
    } else {

        3. _____
    }
}
```

```
# Python
def getHeight(root):
    if root == None:

        1. _____

    l = getHeight(root.left)
    r = getHeight(root.right)

    if l > r:

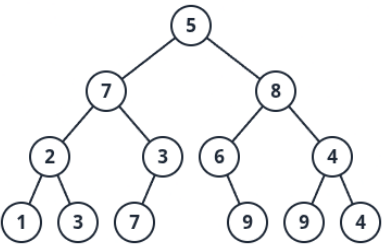
        2. _____
    else:

        3. _____
```

Question#2 [Coding] [12 Marks]

Complete the method **sumOfNode(BTNode root)**, which takes the root of a binary tree as a parameter. The method should compute and **return** the sum of the left child values of nodes at even levels and the right child values of nodes at odd levels. Each node has three instance variables: left (BTNode), right (BTNode) and elem (Object datatype).

Note: The root is at level 0. Do not include the root node's value in the sum. (You may use a helper function if needed)

Sample Given Tree	Returned Value	Explanation
	32	Level 0 is Even, take Left child's value 7. Level 1 is Odd, take Right child's value 3, 4. Level 2 is Even, take Left child's value 1, 7, 9. The sum = 7+3+4+1+7+9 = 32

Java

```
public static int sumOfNodes( BTNode root ){
```

||

Python

||

```
def sumOfNodes( root ):
```